

Neutrinoless Double Beta Decay Transformer Benchmarks (Working Title)

Anonymous authors

Paper under double-blind review

Abstract

The abstract paragraph should be indented 1/2 inch (3 picas) on both left and right-hand margins. Use 10 point type, with a vertical spacing of 11 points. The word Abstract must be centered, in small caps, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 Datasets

This section introduces four rare-event physics experiments and the public datasets released by each collaboration for this benchmark. Each experiment records data from a different kind of particle detector while searching for the same extremely rare, hypothesized signal, one expected to make up only a tiny fraction of everything actually recorded. As shown in Figure 1, the four datasets differ substantially in their native data modality; one-dimensional waveforms, three-dimensional point clouds, hybrid trajectory-and-energy representation, and spatial energy clusters. This makes them a useful and diverse testbed for general-purpose classification architectures.

The Majorana Demonstrator Arnquist et al. (2023a) is built around an array of germanium crystal detectors. The public data release (Arnquist et al., 2023c) consists of calibration-run data in HDF5 format, with each example a single digitized detector waveform 3800 samples long, spanning about 38 microseconds, together with its calibrated energy and detector/run identifiers. The waveform has a simple, recognizable shape: a rising edge followed by a long, slowly decaying tail. The exact shape of that rise carries the classification signal, signal-like waveforms rise quickly and cleanly, while background-like waveforms tend to rise more slowly or in multiple steps. The dataset provides several such shape-based labels per event. NEXT Novella et al. (2023) uses a large gas-filled chamber that reconstructs each event as a three-dimensional shape rather than a simple waveform. The dataset used here (Mistry, 2026) consists of simulated signal and background events chosen because the two classes have very similar total energy, making the task difficult if energy alone is used. Each event is represented as a sparse point cloud on a $3 \times 3 \times 3$ mm³ grid: a variable-length list of (x, y, z, energy) tuples marking where energy was deposited, rather than a fixed-size image or waveform. The key distinguishing feature is geometric: signal-like events form one continuous track with two dense clusters of energy at its two ends, while background-like events typically show only one such cluster. The classification task is binary, based on this 3D structure. SuperNEMO Petro et al. (2025) combines two detector types in one device: a tracking volume that records individual particle trajectories, and a surrounding array of energy-measuring blocks. The dataset used here (Macko, 2026b) consists of simulated events from four different underlying processes, one signal-like class and three background-like classes, making the native classification task four-way rather than binary. Each event combines two structurally different pieces of information: a discrete, graph-like set of tracker hits describing a particle’s path, and a small set of calorimeter readings giving its energy and arrival time. This hybrid representation, part trajectory, part fixed-length energy measurement, sets SuperNEMO’s data apart from the single-modality representations used by the other experiments here. EXO-200 Anton et al. (2019) was a liquid-filled chamber that recorded two parallel signals for every event: an electrical charge signal and a light signal. The public data release (EXO-200 Collaboration, 2026) provides calibration-run data in HDF5 format. Each example includes the two raw waveforms plus several derived quantities: two separate energy estimates, one from each signal, a combined energy value, and a description of how many separate energy deposits were found and where. That last feature is the main classification signal, signal-like events deposit their energy in one tight, localized spot, while

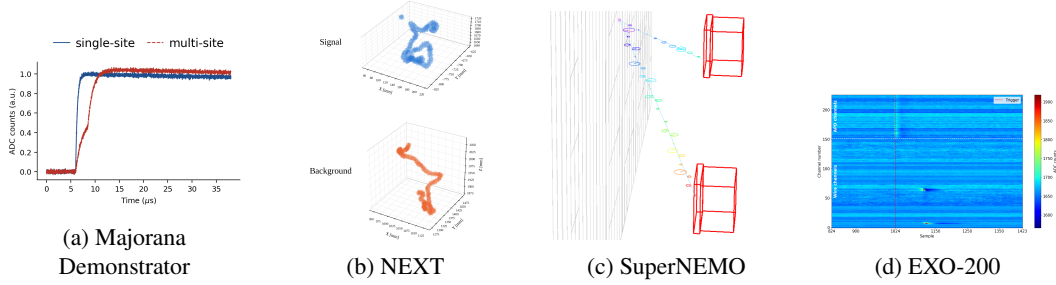


Figure 1: Native data modality of each of the four experiments considered in this benchmark: (a) detector waveform, (b) 3D point cloud, (c) hybrid trajectory-and-energy event display, and (d) spatial energy clusters.

background-like events tend to leave energy in several separated spots. The native classification task is therefore single-cluster versus multi-cluster.

2 Benchmarking Metrics

2.1 The Energy-Bias Problem

Strong classification performance can reflect both event morphology and energy-related differences between classes. Energy is valid physical information, but comparisons intended to assess structure beyond energy should also control the class energy spectra. Otherwise, accessible energy cues can provide a shortcut relative to that objective (Geirhos et al., 2020).

MJD provides a concrete example. Its benchmark distinguishes clean events, which pass all four reference pulse-shape cuts, from non-clean events. Pulse height is approximately related to calibrated energy in the released waveforms (Arnquist et al., 2023c). Our classification preprocessing subtracts the mean of the first 200 samples and divides each waveform by its maximum absolute amplitude. Although this removes the leading amplitude scale, it need not remove energy information from the remaining waveform.

Figure 2 compares low- and high-energy clean events from the same detector and run. Across all 410 clean test events in this group, the median normalized baseline RMS is 7.56 times larger in the lowest than in the highest energy quartile, with 103 events per cohort. The association is also visible across the full group (Spearman $\rho = -0.826$). Thus relative noise can carry an energy cue after amplitude normalization, even with the true class, detector, and run held fixed. The waveform precedes the model-specific representation adapter; this diagnostic does not establish which cues a trained classifier uses.

The clean and non-clean test populations have different energy spectra. Together, these observations motivate evaluating discrimination after balancing class energy distributions and separately testing the energy dependence of predictions. Appendix B gives the selection rule and normalization details. SuperNEMO supplies a complementary example of energy-associated tracker geometry (Appendix D).

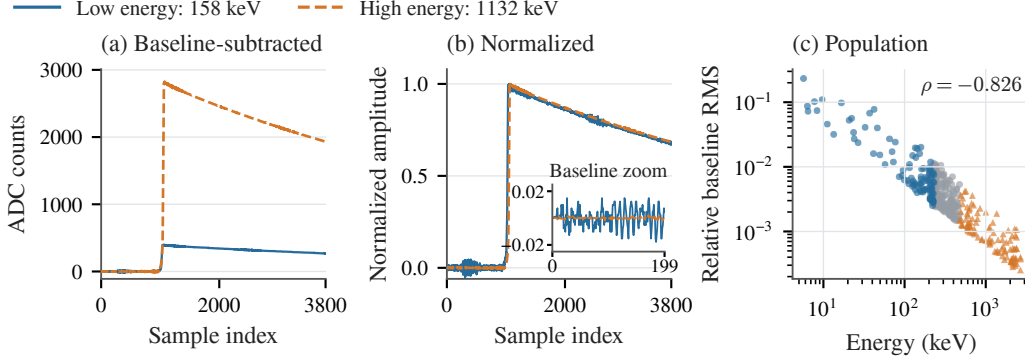
2.2 EnergyBench: Energy-Aware Evaluation

EnergyBench addresses the problem in Section 2.1 with two complementary diagnostics: discrimination after matching energy spectra, and stability of the score distribution with energy within each true class. Inclusive AUC remains useful for performance under the original spectra.

Energy-matched AUC. For event x_i , let $y_i \in \{0, 1\}$, E_i , and s_i denote its evaluation label, conditioning energy, and score, with larger scores favoring $y_i = 1$. Within the classes’ common energy support, EnergyBench reweights events so that both classes assign the same mass to each retained energy bin. The shared target is proportional to the smaller of the two class-normalized bin masses.

MJD waveforms and energy

Clean test events; detector 133, run 50928



(a,b) Examples nearest cohort energy medians. (c) All 410 events: low / high quartiles: 103 each; middle half: 204 (gray). Normalization precedes model-specific representations; RMS uses samples 0–199.

Figure 2: Energy information in MJD waveforms before and after amplitude normalization. The two displayed clean events have energies 157.98 and 1131.51 keV and come from detector 133, run 50928; each is nearest its energy cohort’s median. Baseline-subtracted and normalized waveform panels use the same events, with a baseline zoom. The population panel includes all 410 clean test events from this detector/run, with the lowest and highest energy quartiles highlighted. Relative baseline noise remains associated with energy after normalization. This is a waveform diagnostic, not a measurement of classifier-score dependence.

Writing w_i for the resulting weights, normalized to one per class, the metric is

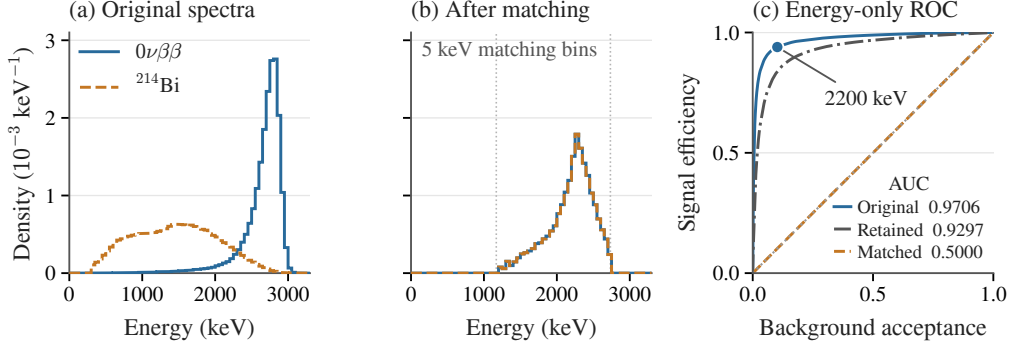
$$\text{AUC}_{\text{match}} = \sum_{i:y_i=1} \sum_{j:y_j=0} w_i w_j [1\{s_i > s_j\} + \frac{1}{2}1\{s_i = s_j\}]. \quad (1)$$

This is one pooled weighted AUC, including cross-bin pairs and half credit for ties (Hanley & McNeil, 1982). Matching changes evaluation weights, without retraining or random resampling. It measures discrimination after balancing bin masses; residual energy differences within bins and exclusion of unsupported energies limit the interpretation. Figure 3(a,b) illustrates the spectra before and after matching for SuperNEMO $0\nu\beta\beta$ versus ^{214}Bi , using the same population and weights as the energy-only ROC in panel (c).

Energy independence score. The second metric asks whether the score distribution changes with energy *within a true class*. For class or physical category g , let P_{gk} and P_g be its normalized score histograms in energy bin k and in the pooled group, and let π_{gk} be the retained bin-weight fraction. Using Jensen–Shannon divergence in natural-log units (Lin, 1991),

$$I_g = 1 - \sqrt{\frac{\sum_k \pi_{gk} \text{JS}(P_{gk}, P_g)}{\ln 2}}, \quad I = \frac{\sum_g A_g I_g}{\sum_g A_g}, \quad (2)$$

where A_g is the finite-pair group base weight within the specified energy range before sparse-bin exclusion, and only estimable groups are averaged. This calculation uses the original event weights. Higher $I \in [0, 1]$ indicates less detected dependence at the chosen histogram resolution; it does not certify statistical independence. Constant predictions attain $I = 1$ but AUC 1/2, so both metrics are needed: matched AUC measures discrimination, while I measures score stability. These metrics are complementary, without implying statistical independence or orthogonality. At the population level, exact score–energy independence within each class makes matched and inclusive AUC equal: energy-only selection and reweighting preserve the class-conditional score distributions. However, the binned estimate $I = 1$ does not establish this independence or guarantee equal AUCs. Conversely, equal AUCs do not imply $I = 1$, since score distributions can vary with energy while preserving signal–background rankings.



$I = 0.079$; retained: 50.4% signal, 57.4% background (original counts).

283 matching bins (5 keV + overflow); spectra displayed in 50 keV bins.

Figure 3: SuperNEMO energy matching and its effect on the fixed score $s = E_1 + E_2$ for $0\nu\beta\beta$ versus ^{214}Bi . (a,b) Class-normalized energy spectra before and after matching, displayed in 50 keV bins. The original sample contains 196,608 signal and 262,144 background events. Matching equalizes class mass in 283 valid 5 keV bins from the shared regular-plus-overflow grid and retains 50.4% and 57.4% of events, respectively; vertical guides in (b) mark the common-support boundaries. (c) Original, unweighted retained-population, and matched AUCs are 0.9706, 0.9297, and 0.5000. The retained and matched curves use the same events after support restriction and sparse-bin exclusion. The point marks the 2200 keV cut on the original population.

Energy-only diagnostic. Figure 3(c) applies the metrics to $s = E_1 + E_2$ for SuperNEMO $0\nu\beta\beta$ versus ^{214}Bi , a separate illustration from its trained $2\nu\beta\beta$ task. Inclusive AUC is 0.9706. Common-support restriction and sparse-bin exclusion retain 50.4% of signal and 57.4% of background events across 283 valid 5 keV bins. On these same retained events, unweighted AUC is 0.9297 and energy-matched AUC is 0.5000. Thus the retained-population curve separates the effect of reweighting from event exclusion. The score-stability diagnostic uses the same regular and overflow energy grid within each class, with up to 20 score-quantile bins, and gives $I = 0.0792$. These calculations illustrate an energy-only decision rule without attributing that strategy to a trained classifier.

Evaluation workflow. Evaluation starts from saved event-level scores, labels, physical energies, and test identifiers. Scores are oriented toward the evaluation-positive class and energies converted to keV. The MJD and EXO-200 classic models, nine EXO-200 Transformers, and six MJD MLP/Fourier Transformers use exactly 600 fixed 5 keV bins with edges 0, 5, ..., 3000 keV. Intervals are left-closed/right-open, except that the last includes 3000 keV. Energies outside 0–3000 keV are excluded without clipping; inclusive AUC retains its original finite-score population regardless of energy. Finite score/energy pairs with positive base weight enter two separate calculations. For I , each group/bin requires at least 20 events; retained events define the weighted score-quantile histograms. For matching, common-support trimming precedes the requirement of at least 20 events per class/bin, followed by overlap reweighting and a pooled AUC. Reporting requires at least 50% coverage of each original finite-pair class base mass and at least two valid bins; only the bin-count requirement has an exception for exactly constant common energy. Exact weights, boundaries, and reporting rules appear in Appendix A. NEXT and SuperNEMO retain their recorded profiles, including an overflow bin for their classic models and the separate energy-only illustration. The three MJD RoPE entries retain their recorded results; they are not part of the MJD reevaluation.

3 Model Design

We compare two approaches to modeling heterogeneous detector data: architectures specialized for each detector modality and a shared Transformer backbone adapted through detector-aware tokenization and positional encoding. The specialized models provide modality-specific inductive biases,

whereas the Transformer experiments test whether a common architecture can operate across detectors through changes to its input representation. Training, checkpoint selection, and the limits of cross-family comparison are described in Section 4.1; evaluation protocols are described in Section 4.2.

3.1 Non-Transformer Models and Selection Rationale

Our NEXT exploration spans 23 recorded non-Transformer classification candidates, organized into four groups according to their input representation and mechanism for aggregating event information (Table 10). Projection and voxel models operate on regular spatial grids, using 2D or 3D convolutions, sparse convolutions, or projection-patch mixing. Point-set models aggregate unordered hits through shared point features and local geometric neighborhoods. Graph models explicitly propagate messages along fixed or learned neighborhood connections. Sequence, state-space, topology and hybrid models explore ordered context accumulation, topological summaries, or combined representations. The complete NEXT inventory uses these same group definitions.

The multi-view CNN applies a shared convolutional encoder to three views and combines their embeddings through late fusion. It captures local patterns while retaining view identity, and the same backbone accepts spatial projections or waveform spectrograms. Static GINE uses edge-aware message passing on a fixed nearest-neighbor graph (Xu et al., 2019). For spatial data, this retains neighborhood relations that projections can obscure; for waveforms, the graph is constructed in pulse-shape coordinates. BiGRU processes two complementary Hilbert orders with bidirectional recurrence (Cho et al., 2014), providing a baseline for aggregating information along ordered event points. PointMamba-lite applies input-dependent state-space updates to the ordered points, providing a different mechanism for accumulating context (Gu & Dao, 2023; Liang et al., 2024). It is our local reduced implementation, rather than a reproduction of the official PointMamba model.

We carry multi-view CNN, Static GINE, BiGRU, and PointMamba-lite into the cross-dataset comparison to represent four complementary architectural inductive biases. The multi-view CNN models regular-grid locality across multiple detector projections; Static GINE represents events through explicit graph connectivity and neighborhood message passing; BiGRU captures bidirectional sequential dependencies; and PointMamba-lite uses selective state-space evolution for long-range sequence modeling. These models were selected to provide broad architectural coverage rather than as winners of a uniform preliminary search. The resulting set is therefore not a one-representative-per-taxonomy-group selection: no point-set architecture is included in the shared cross-dataset campaign. Each architecture is trained separately on every dataset using its detector-specific input adapter and architecture-appropriate optimization configuration. Detailed model structures, selection provenance, and additional architectures considered during development are provided in Appendix E.1. Appendix E and Section 4.

3.2 Tokenized Transformer Baselines

We map each detector event to a common Transformer interface. For detector d and tokenization strategy r , the tokenizer maps event x to

$$\mathcal{T}_{d,r}(x) = (\mathbf{F}, \mathbf{P}, \mathbf{m}),$$

where $\mathbf{F}$ contains token content, $\mathbf{P}$ contains positional or structural descriptors supplied to the positional encoder, and $\mathbf{m}$ identifies valid tokens. A learned projection maps token content into the Transformer embedding space. Separate models are trained for each dataset and representation while the core Transformer dimensions are held fixed.

We compare three tokenization methodologies representing detector data at different levels of granularity. Entity tokens preserve individual measurements: NEXT and SuperNEMO use detector or tracker hits, MJD uses selected waveform samples, and EXO-200 uses sensor-time points balanced across its six sensor blocks. Region tokens retain bounded local structure: NEXT and SuperNEMO aggregate occupied three-dimensional regions, MJD uses non-overlapping waveform patches, and EXO-200 uses regions defined jointly over sensor blocks, channel groups, and time intervals. Summary tokens apply stronger compression by replacing groups of measurements with descriptive statistics: NEXT and SuperNEMO form balanced Morton-ordered spatial groups, MJD summarizes temporal waveform segments, and EXO-200 summarizes sensor-time regions. These alternatives test

whether classification benefits most from preserving individual measurements, retaining local neighborhood structure, or applying statistical compression. Detailed implementations and token budgets appear in Appendix G.

We evaluate learned coordinate-MLP encoding, multiscale Fourier features, and rotary positional encoding (RoPE). The additive MLP and Fourier methods map each token’s positional or structural descriptors into the embedding space, whereas RoPE injects descriptor-dependent rotations into the query and key vectors used by self-attention. The shared backbone uses 64-dimensional embeddings, two Transformer encoder layers, four attention heads, feed-forward width 256, and masked mean pooling followed by a binary classification head. Input projections and positional mechanisms change the parameter count, and representation-dependent sequence lengths change the computational cost even though the core backbone dimensions remain fixed.

4 Experiment Results

4.1 Experimental Setup

To ensure comparable results, all architectures within each dataset use the same classification target, recorded data partition, and held-out evaluation protocol. We additionally fix the optimization procedure and Transformer backbone across all tokenization and positional-encoding variants, allowing their effects to be studied without changing the underlying architecture. The specialized baselines retain their recorded architecture-specific training configurations, but are evaluated on the same test events and with the same metrics as the Transformers. The benchmark therefore provides a controlled comparison of representations and model families, while not claiming an identical computational budget across architectures.

The stored NEXT labels assign 1 to $0\nu\beta\beta$ and 0 to ^{214}Bi , while SuperNEMO assigns 1 to $2\nu\beta\beta$ and 0 to ^{214}Bi . EXO-200 stores label 1 for events with more than one charge cluster and label 0 for single-cluster events; during evaluation, we reverse the label and score orientation so that single-cluster events form the positive class, following Table 2. MJD assigns label 1 only when all four reference pulse-shape-discrimination criteria pass. The evaluation-positive class therefore has a dataset-specific physical interpretation. NEXT uses an approximately 80/10/10 training/validation/test partition based on class-wise event counts, and different portions of a source file may enter different splits. SuperNEMO uses the same approximate proportions while assigning blocks of complete events within each source category. EXO-200 keeps acquisition runs disjoint across splits. MJD preserves its official test files and divides the provided training data 90/10 into training and validation subsets. Each model for a given dataset reuses the same recorded partition.

All Transformer baselines use batch size 64, at most 50 epochs, initial learning rate 5×10^{-4} , weight decay 10^{-4} , gradient-norm clipping at 1, early-stopping patience 5, and random seed 42. We optimize binary cross-entropy on logits with AdamW and a cosine learning-rate schedule, and select checkpoints by validation inclusive AUC. The audited neural specialized baselines use the same general loss and optimization family, but the saved runs retain architecture-specific batch sizes, learning rates, and early-stopping settings, summarized in Table 9. The boosted-tree baseline instead uses its recorded estimator configuration (Appendix E). We therefore report the specialized models as recorded rather than claiming an identical training budget across all architectures. Finally, the saved SuperNEMO non-Transformer checkpoints record selection by validation inclusive AUC, although the current training implementation selects energy-matched AUC; throughout this work, we treat the saved checkpoint metadata as the authoritative description of the reported runs.

4.2 Classification Results

Table 1 reports matched AUC, energy independence I , and trainable parameter counts. Within each dataset, all models use the same metric definitions and evaluation protocol, allowing direct comparison between specialized baselines and tokenized Transformers. Matched AUC measures discrimination after balancing the class energy spectra, while I measures within-class score stability with energy.

Table 1: Cross-detector classification results. Higher values are better for energy-matched AUC (AUC_{match}) and energy independence (I). Parameters are rounded counts or ranges in thousands across available dataset-specific implementations; ranges are not uncertainty. Within each model family and dataset, bold marks the highest eligible result where recorded precision resolves the comparison; displayed ties use unrounded values where available. Blue marks the highest AUC and I across both model families in each dataset. Highlighting denotes point estimates, not statistical significance. Results are from single training seeds. A dash denotes an unavailable result. [‡]MJD GINE uses additional PSD-label supervision and is excluded from best-result highlighting. Evaluation details are provided in Appendix A.

Model	#Params (K)	NEXT		MJD		EXO-200		SuperNEMO	
		AUC _{match}	<i>I</i>	AUC _{match}	<i>I</i>	AUC _{match}	<i>I</i>	AUC _{match}	<i>I</i>
<i>Specialized baselines</i>									
Multi-view CNN	808	0.977	0.971	0.970	0.760	0.863	0.752	0.586	0.910
Static GINE	480	0.995	0.970	0.969 [‡]	0.797 [‡]	0.949	0.801	0.590	0.913
BiGRU	734	0.979	0.971	0.975	0.775	0.929	0.789	0.579	0.921
PointMamba-lite	317	0.925	0.971	0.960	0.773	0.934	0.786	0.579	0.922
<i>Tokenized Transformers</i>									
Entity + MLP	111	0.975	0.975	0.929	0.812	0.941	0.737	0.591	0.975
Entity + Fourier	113–116	0.963	0.974	0.941	0.802	0.947	0.729	0.589	0.974
Entity + RoPE	106–113	0.497	0.512	0.793	0.749	0.946	0.715	0.551	0.865
Region + MLP	111–121	0.988	0.973	0.973	0.798	0.970	0.801	0.591	0.976
Region + Fourier	113–125	0.979	0.973	0.981	0.788	0.972	0.790	0.590	0.977
Region + RoPE	107–116	0.512	0.483	0.916	0.768	0.971	0.789	0.602	0.953
Summary + MLP	111	0.987	0.971	0.965	0.799	0.947	0.776	0.590	0.975
Summary + Fourier	113–116	0.983	0.973	0.969	0.782	0.960	0.802	0.589	0.977
Summary + RoPE	106–114	0.602	0.532	0.878	0.777	0.955	0.773	0.582	0.956

Specialized-baseline comparison. Static GINE has the highest matched-AUC point estimate on NEXT, EXO-200, and SuperNEMO among the shared classic models. It also has the highest I on EXO-200, where the recalculated values are 0.949 and 0.801, respectively. BiGRU has the highest matched AUC on MJD, while GINE has the highest I with additional PSD-label supervision. These matched-AUC leaders are unchanged by reevaluation. On SuperNEMO, PointMamba-lite has the highest I and now marginally exceeds BiGRU in matched AUC (0.579492745 versus 0.579485174); this small ordering reversal removes BiGRU from the two-metric Pareto set, leaving GINE and PointMamba-lite. These single-run point estimates do not establish statistical significance. Parameter counts and training protocols also differ, including the additional PSD supervision for MJD GINE.

The finer common energy grid lowers I for the EXO-200 and SuperNEMO classic models relative to their historical quantile-energy estimates; both class-level values decline, as documented in Table 5. This reflects a changed estimator resolution and pooling rule, not a change in the fitted models. All reported retained-campaign classic matched AUCs meet the original count and coverage gates. Coverage, overflow counts, sparse-bin losses, and effective sample sizes are reported in Appendix A.

Among the recorded Transformer results, region tokens provide the strongest matched-AUC configuration on each detector: region-MLP on NEXT, region-Fourier on MJD and EXO-200, and region-RoPE on SuperNEMO. Summary tokens remain close to the leading NEXT representation, suggesting that spatial aggregation can preserve much of the useful event structure. Entity tokens are generally weaker on the dense MJD and EXO-200 waveforms, where selecting isolated measurements can discard contiguous pulse structure retained by regional patches. Across the completed MLP and Fourier comparisons, changing tokenization generally produces a larger performance range than changing positional encoding within one tokenization. This supports the conclusion that defining what constitutes a token is a central modeling decision, although the evidence consists of single-seed results.

Effect of positional encoding and energy dependence. Positional-encoding preferences remain detector dependent. Coordinate MLPs perform best across the NEXT tokenizations, whereas Fourier features provide the strongest MLP/Fourier results on MJD and EXO-200. SuperNEMO shows smaller differences among the additive encodings, with its reported region-RoPE variant achieving the highest Transformer matched AUC. NEXT and MJD are more sensitive to RoPE, but for different reasons: NEXT supplies physical spatial coordinates, while MJD supplies time together with waveform level and local change. The comparison therefore varies both coordinate semantics and the encoding mechanism, so it does not isolate a universal positional-encoding effect.

Within model subsets evaluated under a common protocol, the strongest classifier is not always the most energy independent. SuperNEMO further shows that high independence can coexist with weak class separation. Energy-matched AUC and I must therefore be interpreted jointly: the former measures discrimination after energy-spectrum balancing, whereas the latter measures within-class score stability with energy.

5 Conclusion

Figure 4 compares model capacity, energy-matched ranking error, and energy independence for 19 NEXT neural configurations. The complete results are listed in Table 10.

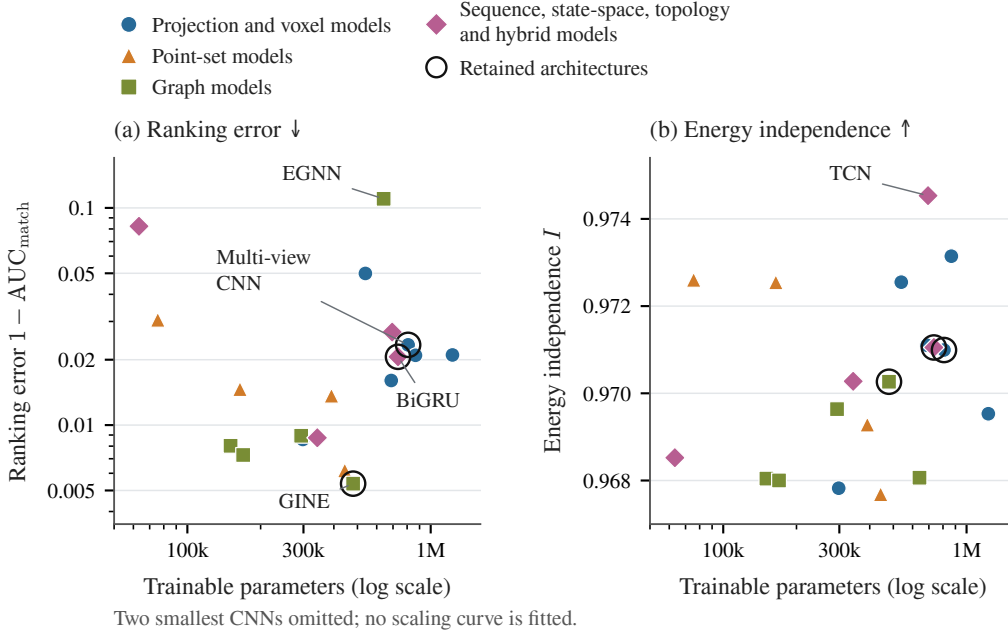


Figure 4: NEXT capacity comparison for 19 neural configurations on 116,549 test events (the shared 5 keV plus overflow protocol), excluding Two-conv CNN and Energy-skip CNN. (a) Weighted pairwise ranking error $1 - \text{AUC}_{\text{match}}$. (b) Energy independence I . Parameter count and ranking error use log scales; I uses a linear scale. Colors and shapes identify four model groups; black rings mark the three retained architectures with campaign results. Earlier exploratory runs are excluded. The full inventory is in Table 10. No scaling curve is fitted.

The shared classic-model reevaluation separates energy-matched ranking from within-class score stability. The matched-AUC leaders among the available retained-campaign classic models remain unchanged, while the independence values on EXO-200 and SuperNEMO are lower under the finer common grid. The MJD and EXO-200 reevaluation excludes energies outside 0–3000 keV without changing inclusive AUC. The reevaluated Transformers retain the same matched-AUC leaders, while EXO-200 region-RoPE leaves the Transformer Pareto set after its independence score decreases under the fixed grid. These single-run comparisons describe the retained test populations and do not establish statistical significance or performance under distribution shift.

AI use statement

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, “we checked LLM-generated research ideas for potential plagiarism through a manual literature survey”, “LLM-generated code was verified and tested for correctness by 2 authors”, etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

See the ICLR 2027 AI Policy for Authors for more details. This statement should not be more than 1 page.

Ethics statement

(This section is **recommended** and does not count toward the page limit.)

If authors feel that their paper submission raises questions regarding the Code of Ethics, they are encouraged to include a paragraph of Ethics Statement (at the end of the main text before references) to address potential concerns where appropriate. Topics include, but are not limited to, studies that involve human subjects, practices to data set releases, potentially harmful insights, methodologies and applications, potential conflicts of interest and sponsorship, discrimination/bias/fairness concerns, privacy and security issues, legal compliance, and research integrity issues (e.g., IRB, documentation, research ethics). This statement should not be more than 1 page.

Reproducibility statement

(This section is **recommended** and does not count toward the page limit.)

It is important that the work published in ICLR is reproducible. Authors are strongly encouraged to include a paragraph-long Reproducibility Statement at the end of the main text (before references) to discuss the efforts that have been made to ensure reproducibility. This paragraph should not itself describe details needed for reproducing the results, but rather reference the parts of the main paper, appendix, and supplemental materials that will help with reproducibility. For example, for novel models or algorithms, a link to an anonymous downloadable source code can be submitted as supplementary materials; for theoretical results, clear explanations of any assumptions and a complete proof of the claims can be included in the appendix; for any datasets used in the experiments, a complete description of the data processing steps can be provided in the supplementary materials. Each of the above are examples of things that can be referenced in the reproducibility statement.

Author Contributions

If you’d like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

Acknowledgments

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

References

- G. Anton et al. Search for neutrinoless double- β decay with the complete EXO-200 dataset. *Physical Review Letters*, 123:161802, 2019.
- I. J. Arnuist et al. Final result of the MAJORANA DEMONSTRATOR’s search for neutrinoless double- β decay in ^{76}Ge . *Physical Review Letters*, 130:062501, 2023a.

- I. J. Arnquist et al. Majorana Demonstrator data release for AI/ML applications, 2023b. URL <https://arxiv.org/abs/2308.10856>.
- I. J. Arnquist et al. Majorana Demonstrator Data Release for AI/ML Applications. *arXiv preprint arXiv:2308.10856*, 2023c.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, pp. 1724–1734, 2014. doi: 10.3115/v1/D14-1179. URL <https://aclanthology.org/D14-1179/>.
- EXO-200 Collaboration. EXO-200 Public Data Release for AI/ML Applications, 2026. [Dataset]. <https://doi.org/10.5281/zenodo.22155860>.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020. doi: 10.1038/s42256-020-00257-z. URL <https://www.nature.com/articles/s42256-020-00257-z>.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023. URL <https://arxiv.org/abs/2312.00752>.
- James A. Hanley and Barbara J. McNeil. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1):29–36, 1982. doi: 10.1148/radiology.143.1.7063747. URL https://www.med.mcgill.ca/epidemiology/hanley/software/hanley_mcneil_radiology_82.pdf.
- Dingkang Liang, Xin Zhou, Wei Xu, Xingkui Zhu, Zhikang Zou, Xiaoqing Ye, Xiao Tan, and Xiang Bai. Pointmamba: A simple state space model for point cloud analysis. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/395371f778ebd4854b88521100af30ad-Abstract-Conference.html.
- Jianhua Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991. doi: 10.1109/18.61115. URL <https://ieeexplore.ieee.org/document/61115/>.
- Miroslav Macko. SuperNEMO simulation dataset for $0\nu\beta\beta$ AI summer school at University of California, Irvine 2026, June 2026a. Dataset release documentation, 18 June 2026, on behalf of the SuperNEMO Collaboration.
- Miroslav Macko. SuperNEMO Dataset for $0\nu\beta\beta$ AI Summer School - UC Irvine, USA (20-21 June 2026), 2026b. [Dataset]. <https://doi.org/10.5281/zenodo.20698789>.
- Krishan Mistry. NEXT Simulation Dataset for AI Summer School UC Irvine 2026. *arXiv preprint arXiv:2605.07566*, 2026.
- P. Novella et al. Demonstration of neutrinoless double beta decay searches in gaseous xenon with NEXT. *Journal of High Energy Physics*, 09:190, 2023.
- M. Petro et al. Status of the SuperNEMO experiment and sensitivity estimates to $0\nu\beta\beta$ and beyond. *PoS, MEDEX2025:034*, 2025.
- Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? *arXiv preprint arXiv:1810.00826*, 2019. URL <https://arxiv.org/abs/1810.00826>.

A Full Metric Definitions and Evaluation Protocols

The 0–3000 keV protocol below applies to the reevaluated MJD and EXO-200 classic models, nine EXO-200 Transformer variants, and six MJD Transformer variants with MLP or Fourier position encoding. The NEXT and SuperNEMO classic results and the separate SuperNEMO energy-only illustration retain their preceding evaluation profile, specified below. The three MJD RoPE variants and the NEXT and SuperNEMO Transformer entries retain their historical reported values outside this reevaluation.

A.1 Energy-matched AUC

For event x_i , let $y_i \in \{0, 1\}$ denote its evaluation label, E_i its conditioning energy, s_i its scalar classifier score, and $a_i \geq 0$ its base event weight. Scores are oriented so that larger values favor $y_i = 1$; this convention need not coincide with the stored training label (Table 2). The selected evaluations use unit base weights. We first describe the weighted form implemented by the evaluators. Only finite scores with positive base weight contribute to inclusive AUC; energy-dependent metrics additionally require finite physical energy satisfying $0 \leq E_i \leq 3000$ keV. This range restriction precedes support estimation.

The common support $C = [L, U]$ is the intersection of the two classes’ central energy ranges:

$$L = \max_{c \in \{0,1\}} Q_c(\alpha), \quad U = \min_{c \in \{0,1\}} Q_c(1 - \alpha), \quad \alpha = 0.005, \quad (3)$$

where Q_c is the base-weighted empirical energy quantile of finite-score, finite-energy, positive-weight events in class c within 0–3000 keV. No additional matching region of interest is specified in these runs. For energy bins B_k , define the count n_{ck} and base mass m_{ck} of class- c events in $C \cap B_k$, and let $p_{ck} = m_{ck} / \sum_{\ell} m_{c\ell}$. A bin is retained only if it contains at least 20 positive-weight events from each class. Writing $\mathcal{V}$ for these bins, the implemented overlap target and event weights are

$$q_k = \frac{1\{k \in \mathcal{V}\} \min(p_{0k}, p_{1k})}{\sum_{\ell \in \mathcal{V}} \min(p_{0\ell}, p_{1\ell})}, \quad (4)$$

$$w_i = a_i \frac{q_k}{m_{y_i k}} \quad \text{if } E_i \in C \cap B_k, \quad k \in \mathcal{V}; \quad w_i = 0 \text{ otherwise.} \quad (5)$$

Thus each class has unit total matched weight and assigns mass q_k to bin k . The target favors energy regions represented in both classes and is defined by the minimum of their class-normalized bin masses.

All retained events are pooled into one weighted ROC curve. Equivalently, with $H(u) = 1\{u > 0\} + \frac{1}{2}1\{u = 0\}$,

$$\text{AUC}_{\text{match}} = \frac{\sum_{i:y_i=1} \sum_{j:y_j=0} w_i w_j H(s_i - s_j)}{\left(\sum_{i:y_i=1} w_i \right) \left(\sum_{j:y_j=0} w_j \right)}. \quad (6)$$

This is the weighted pairwise ranking interpretation of AUC (Hanley & McNeil, 1982); equal scores receive half credit. The code implements the equivalent weighted pairwise calculation by sorting scores and processing ties together. It does not average within-bin AUCs: pairs from different energy bins also contribute to Equation 6. Inclusive AUC uses the same ranking calculation with base weights and without energy restriction.

Matching is a deterministic evaluation operation on the supplied split; it does not resample events or alter the fitted classifier. For the reported test results, support and weights are estimated on the test events themselves. The formal matched AUC is withheld if either class retains less than 50% of its original finite-score/finite-energy class base mass, or fewer than two bins survive. Exactly constant common energy exempts only the two-bin requirement; the per-class event-count and coverage gates still apply. Empty, single-class, and insufficient-count bins receive zero matched weight.

Both energy metrics in the MJD and EXO-200 reevaluation use the same global grid after conversion to keV:

$$B_k = [5k, 5(k+1)) \text{ keV}, \quad k = 0, \dots, 598, \quad (7)$$

$$B_{599} = [2995, 3000] \text{ keV}. \quad (8)$$

There are exactly 600 energy bins. Exactly 3000 keV belongs to the last bin; energies below zero or above 3000 keV are excluded and reported separately. Energies are never clipped. Matching additionally requires $L \leq E_i \leq U$, so support boundaries may cut through an energy bin.

Matching equalizes bin masses, not continuous energy distributions within bins. Finite width, finite statistics, trimming, and sparse-bin exclusion limit the interpretation to the retained support and resolution. A score that is solely a function of the matched bin index has AUC 1/2, because its two weighted class distributions are identical; a score depending on within-bin energy need not. This is a property of the weighting construction, not an empirical claim about the trained models.

A.2 Energy independence score

Energy matching measures discrimination after balancing class spectra. The complementary independence score asks whether the *distribution of classifier scores* changes with energy within a true class. Conditioning on class avoids interpreting class separation itself as unwanted dependence. The evaluator groups by supplied physical category, falling back to the binary label when category metadata are absent. Here each reported task has two such groups. This calculation uses the original base weights, without matching or common-support trimming.

For group g , let P_{gk} be the normalized score histogram in an eligible energy bin, P_g the group's pooled score histogram, and π_{gk} the fraction of retained base weight in that energy bin. Score bins are shared within a group and obtained from up to 20 weighted score-quantile bins. Using natural logarithms, define

$$\text{JS}(P, Q) = \frac{1}{2} \sum_r P_r \log \frac{P_r}{M_r} + \frac{1}{2} \sum_r Q_r \log \frac{Q_r}{M_r}, \quad M_r = \frac{1}{2}(P_r + Q_r), \quad (9)$$

$$D_g = \sum_k \pi_{gk} \text{JS}(P_{gk}, P_g), \quad I_g = \text{clip}_{[0,1]} \left(1 - \sqrt{D_g / \log 2} \right), \quad (10)$$

$$I = \frac{\sum_{g \in \mathcal{G}_*} A_g I_g}{\sum_{g \in \mathcal{G}_*} A_g}. \quad (11)$$

Here r indexes score-histogram bins, zero-probability terms contribute zero, $\mathcal{G}_*$ contains estimable groups, and A_g is their base weight after finite-pair and 0–3000 keV range filtering, but before sparse-bin exclusion. These MJD and EXO-200 evaluations use the retained-event pooling and eligibility rules specified below. The Jensen–Shannon divergence is bounded by $\log 2$ (Lin, 1991), giving $I_g, I \in [0, 1]$ whenever estimable. Larger values indicate less detected dependence at the chosen binning resolution. The report also retains $\min_g I_g$ to expose a poorly behaving group that the weighted mean could obscure.

This is a histogram-based dependence proxy, not Pearson correlation, distance correlation, mutual information, or a test certifying statistical independence. Its ideal value is one when all compared histograms coincide. Constant predictions also attain one while providing AUC 1/2; a single populated energy bin makes the score uninformative about energy dependence. Finite samples can produce nonzero histogram divergences even under population independence, and coarse bins can hide dependence. The score depends on energy coverage, score discretization, and group composition, so it should be compared under a common protocol. No threshold or operating-point diagnostic enters Equations 9–11.

Joint interpretation. High matched AUC indicates ranking ability after the specified spectral balancing; high I indicates stable within-class score distributions across the inspected energy bins. Neither implies the other. Inclusive AUC supplies the complementary performance under the original sampled spectra. Its change after matching is descriptive, rather than a causal decomposition of energy use: both the support and the event weights can change. We report the three quantities together.

Table 2: Conditioning variables for classic models and the reevaluated EXO-200 and MJD Transformers. The MJD and EXO-200 reevaluation uses 600 fixed 5 keV bins on 0–3000 keV and excludes energies outside this range (v3). NEXT and SuperNEMO classic results retain 600 regular bins plus one overflow bin (v2). The two metrics retain distinct statistical definitions.

Dataset	Evaluation-positive class	Physical conditioning energy
NEXT	$0\nu\beta\beta$	Summed deposited-hit energy; MeV converted to keV
MJD	Clean pulse	Original calibrated float64 energy (keV), recovered from test-event metadata
EXO-200	Stored label 0 ($n_{\text{CCL}} = 1$)	Reconstructed rotated energy (keV); negate background-oriented logit
SuperNEMO	$2\nu\beta\beta$	Reconstructed calorimeter $E_1 + E_2$ (keV)

A.3 Implementation Details

Evaluation inputs and orientation. The MJD and EXO-200 reevaluation reuses saved test predictions without training or checkpoint substitution. The unchanged NEXT classic results retain stored scalar classifier scores and convert summed deposited-hit energy from MeV to keV. MJD uses calibrated physical energy recovered in official test-shard/row order. Event identities and the saved labels and energies are verified against the original metadata. The classic adapter’s clipping is removed; Transformer energy caches are restored from float32 to the original float64 values. The three classic binary classifiers and all six reevaluated Transformers retain their single clean-oriented scores, while classic GINE retains the recorded aggregation of four PSD outputs (Appendix E). EXO-200’s stored label 0 is positive ($n_{\text{CCL}} = 1$): we set $y = 1 - y_{\text{stored}}$ and negate the stored background-oriented logit for both metrics, including all nine reevaluated Transformers. The unchanged SuperNEMO classic logits favor $2\nu\beta\beta$ and use reconstructed calorimeter $E_1 + E_2$ in keV. The separate $0\nu\beta\beta$ illustration uses $s = E_1 + E_2$ directly. All selected files have unit base weights. No sigmoid is added to scalar logits: reproducing I requires retaining the score representation as well as the ranking direction.

Exact weighted quantiles. The evaluator sorts finite observations $z_{(j)}$ with positive weights $b_{(j)}$ and assigns cumulative center positions

$$t_j = \frac{\sum_{h \leq j} b_{(h)} - b_{(j)}/2}{\sum_h b_{(h)}}. \quad (12)$$

The empirical quantile $Q(u)$ is linearly interpolated between $(t_j, z_{(j)})$, with endpoint values used outside $[t_1, t_n]$. Stable sorting preserves input order for tied observations. These are the quantiles in Equation 3 and in the histogram construction below. Setting the trim fraction to zero would use each class’s minimum and maximum; the reported configurations instead use 0.005.

Units, boundaries, and population. Physical energy is converted to float64 keV before bin assignment. To avoid a unit-dependent one-step floating-point shift at a theoretical edge, an input MeV value exactly equal to a canonical keV edge divided by 1000 maps to that exact edge. Other nearby observations are not snapped by a tolerance. Internal boundaries enter the bin starting at that boundary, with the explicit 3000 keV exception in Equation 8. Original energies are retained in the audit, including MJD’s 67 events above 3000 keV (66 nonclean and one clean), excluded from the two energy metrics rather than clipped into the 2995–3000 keV bin. Inclusive AUC always uses the original finite-score, positive-weight population, regardless of energy.

Matching calculation and diagnostics. Reproduction first retains finite score/energy pairs with $0 \leq E_i \leq 3000$ keV and positive base weight, then estimates C and assigns the shared 600 bins. Event counts and base masses are computed within $C \cap B_k$. The remaining steps reject bins with fewer than 20 events of either class, compute Equations 4–5, and evaluate one pooled weighted ROC. The class fractions p_{ck} are normalized over all common-support bins *before* invalid bins are removed. Sparse bins are not merged. The formal result is unavailable if support is disjoint, statistics are inadequate,

or a reporting gate fails. Coverage for class c is

$$\gamma_c = \frac{\sum_{i: y_i=c, w_i>0} a_i}{\sum_{i: y_i=c, s_i, E_i \text{ finite}, a_i>0} a_i}. \quad (13)$$

This is retained *base* mass, not the matched mass, which is normalized to one in each class. Its denominator precedes both lower and upper energy range exclusions; range filtering cannot inflate the reported coverage. Range loss, subsequent support loss and sparse-bin loss are recorded separately. Both $\gamma_c \geq 0.5$ and at least two valid bins are required. Exactly constant common energy exempts only the two-bin requirement; the coverage and per-class event-count gates still apply. A diagnostic weighted AUC may be saved even when the formal reporting gate fails; it is not substituted for a missing formal result in our table. The effective sample size for any weights v_i is $(\sum_i v_i)^2 / \sum_i v_i^2$. Table 3 reports support and coverage for the reference CNN prediction sets.

Table 3: Matching diagnostics for the multi-view CNN prediction sets under their dataset-specific profiles (Table 2). All reevaluated classic models within each displayed test population use identical test event IDs, labels, physical energies, and base weights. Coverage γ_c uses the original finite score/energy class base mass before energy-range exclusion; ESS uses matched weights.

Dataset	Common support (keV)	Bins	γ_1	γ_0	ESS ₁	ESS ₀
NEXT	[2434.84, 2480.35]	11	0.9882	0.9700	55195	42260
MJD	[18.15, 2613.79]	493	0.9874	0.9631	141205	213947
EXO-200	[538.18, 2662.57]	426	0.9457	0.9414	51924	61627
SuperNEMO	[350.21, 2218.25]	374	0.9900	0.8988	312506	207113

Table 4: Disjoint matching losses and the separate independence sparse-bin loss, reported as positive/negative event counts. N_1/N_0 is the original test population. Range loss precedes support estimation, then matching sparse-bin exclusion. For MJD and EXO-200, range loss includes energies below 0 or above 3000 keV; the $E > 3000$ column is a subset of that loss. For retained NEXT and SuperNEMO results, only negative energies are excluded and $E > 3000$ counts are retained overflow diagnostics. The separate I sparse loss is measured after the applicable range filtering. Counts are shared by reevaluated classic models within each displayed test population.

Dataset	N_1/N_0	Range loss	$E > 3000$	I sparse loss	Support loss	Matching sparse loss
NEXT	64879/51670	0/0	0/0	0/0	767/1550	0/0
MJD	148251/241749	1/66	1/66	137/214	1746/7378	121/1481
EXO-200	66062/74321	3/1423	3/1423	152/257	3587/2932	0/0
SuperNEMO	327680/262144	0/0	0/40	333/271	3275/26530	0/0

No resampling, new model fitting, bootstrap interval, or multiple-seed average is introduced. Support restriction and matching weights can both alter the AUC relative to the inclusive population; their difference is descriptive, not a causal attribution to energy.

Fixed-grid independence score. The reevaluated MJD and EXO-200 records group events by physical category when supplied, otherwise by binary label. Within each group the evaluator removes nonfinite score/energy pairs, energies outside 0–3000 keV, and zero-weight events, retains only energy bins with at least 20 events, and computes weighted score quantiles using the retained events. Duplicate score edges are removed. For nonconstant scores the extreme edges are moved outward by one floating-point step; constant scores receive one interval centered at that value with half-width $\max(|s|10^{-9}, 10^{-12})$. Score histograms use left-closed/right-open bins with the last right edge included. They are normalized probability masses, not densities. No pseudocount is added.

Let R_g be the retained events, R_{gk} those in bin k , and S_{gr} score-histogram interval r . Then

$$P_{gk,r} = \frac{\sum_{i \in R_{gk}} a_i 1\{s_i \in S_{gr}\}}{\sum_{i \in R_{gk}} a_i}, \quad P_{g,r} = \frac{\sum_{i \in R_g} a_i 1\{s_i \in S_{gr}\}}{\sum_{i \in R_g} a_i}, \quad (14)$$

$$\pi_{gk} = \frac{\sum_{i \in R_{gk}} a_i}{\sum_{i \in R_g} a_i}. \quad (15)$$

Equations 9–10 operate on these histograms. The final group weight A_g includes all finite-pair, positive-weight events in that group within 0–3000 keV, including events from rejected sparse bins. Thus the overall score is not an unweighted class mean and not a mean weighted only by retained events. A group with fewer than 20 finite events or no eligible energy bin is excluded; if no group is estimable, the score is missing. One eligible bin is sufficient for this implementation, unlike the usual two-bin reporting gate for matched AUC.

Table 5: Class-level independence diagnostics under the profiles in Table 2. Retained fractions divide eligible-bin base mass by original finite score/energy class base mass, before energy-range exclusion. Score histograms retain up to 20 weighted score-quantile bins. The audit files include all extended NEXT configurations and full precision.

Dataset / model	I_1	I_0	$\min_g I_g$	Bins _{1/0}	Retained _{1/0}
NEXT / Multi-view CNN	0.9763	0.9643	0.9643	11/11	1.0000/1.0000
NEXT / Static GINE	0.9778	0.9609	0.9609	11/11	1.0000/1.0000
NEXT / BiGRU	0.9753	0.9657	0.9657	11/11	1.0000/1.0000
NEXT / PointMamba-lite	0.9776	0.9632	0.9632	11/11	1.0000/1.0000
MJD / Multi-view CNN	0.7325	0.7763	0.7325	496/524	0.9991/0.9988
MJD / Static GINE	0.8057	0.7911	0.7911	496/524	0.9991/0.9988
MJD / BiGRU	0.7791	0.7730	0.7730	496/524	0.9991/0.9988
MJD / PointMamba-lite	0.7519	0.7865	0.7519	496/524	0.9991/0.9988
EXO-200 / Multi-view CNN	0.7315	0.7709	0.7315	438/484	0.9977/0.9774
EXO-200 / Static GINE	0.7925	0.8093	0.7925	438/484	0.9977/0.9774
EXO-200 / BiGRU	0.7739	0.8033	0.7739	438/484	0.9977/0.9774
EXO-200 / PointMamba-lite	0.7854	0.7863	0.7854	438/484	0.9977/0.9774
SuperNEMO / Multi-view CNN	0.9213	0.8962	0.8962	417/514	0.9990/0.9990
SuperNEMO / Static GINE	0.9241	0.8981	0.8981	417/514	0.9990/0.9990
SuperNEMO / BiGRU	0.9292	0.9105	0.9105	417/514	0.9990/0.9990
SuperNEMO / PointMamba-lite	0.9292	0.9119	0.9119	417/514	0.9990/0.9990

Changes from historical implementations. The historical EXO-200 evaluations, including the nine Transformer variants, used six class-balanced pooled energy-quantile bins for matching, and a separate eight per-category energy-quantile construction for I . Both energy constructions are replaced by Equation 8; the up-to-20 score-quantile bins remain. Their old I implementation could compare a local energy bin containing only two events and pooled all finite events. The common reference instead requires 20 events per group/bin and computes score edges and the pooled histogram only on retained events. Bin weights use retained group mass; final group weights use eligible population mass before sparse exclusion. These choices follow the former NEXT/MJD fixed-grid estimator and preserve the JS-then-square-root and final group-aggregation definitions. The six MJD Transformer caches used 836 fixed 5 keV bins extending to 4180 keV, without clipping. They now use Equation 8 and the same range exclusions as the other reevaluated models. Their original float64 physical energies are restored before binning. These changes can alter both score-histogram populations and support quantiles, even though the historical MJD Transformer bin width was already 5 keV.

Retained NEXT and SuperNEMO profile. The previously reported NEXT and SuperNEMO classic results, including NEXT’s separate exploratory batches, and the SuperNEMO energy-only illustration retain EnergyBench-unified-5keV-overflow-v2.0.0. This profile uses the 600 regular

Table 6: Diagnostics for the 15 reevaluated MJD and EXO-200 Transformers using 600 fixed 5 keV bins on 0–3000 keV. All match the CNN test identities, class partitions, physical energies and unit weights within their dataset, so common support and matching losses equal Tables 3–4. Retained fractions and coverage use original finite-pair class mass before range exclusion. The minimum group score is the smaller of the two displayed I_g values. All formal matched AUCs pass the reporting gates.

Dataset / Transformer	I_1/I_0	Bins $_{I,1/0}$	Retained $_{I,1/0}$	K_{match}	γ_1/γ_0	ESS $_{1/0}$
EXO-200 / entity-mlp	0.6868/0.7828	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / entity-fourier	0.6545/0.7969	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / entity-rope	0.6583/0.7671	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / region-mlp	0.7893/0.8115	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / region-fourier	0.7646/0.8131	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / region-rope	0.7502/0.8246	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / summary-mlp	0.7296/0.8183	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / summary-fourier	0.7894/0.8132	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / summary-rope	0.7713/0.7740	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
MJD / entity-mlp	0.8388/0.7950	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / entity-fourier	0.8196/0.7913	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / region-mlp	0.7935/0.8013	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / region-fourier	0.7869/0.7881	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / summary-mlp	0.8019/0.7973	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / summary-fourier	0.7914/0.7759	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947

bins in Equation 8 plus $B_{600} = (3000, \infty)$ keV. Negative energies are excluded, while all finite nonnegative energies enter the support-quantile calculation and the pre-sparse group masses A_g . The overflow bin is subject to the same event-count rules as a regular bin. The score-quantile histograms, overlap target, weighted AUC and original finite-pair coverage denominators use the same definitions as above. These retained results and their figures are not recomputed under the MJD/EXO-200 range restriction.

Reproduction and validation. The MJD/EXO-200 configuration is EnergyBench-unified-5keV-range-v3.0.0. The preceding v2 configuration remains available for the retained NEXT and SuperNEMO results; each result records its applicable profile and SHA256 fingerprint. The reproducibility bundle contains the corresponding frozen evaluators, event-input manifests and hashes, full-precision metrics, class-level bin counts and losses, and a table generator that reads these results. Numerical checks cover unit equivalence at all 601 finite edges, adjacent floating-point values, exclusion outside 0–3000 keV, constants, ties, missing and sparse inputs, weighted pair enumeration, and replay of the historical fixed-grid estimator on identical supported inputs. Independent recalculation checks the metrics, coverage gates, and original event sources.

Degeneracies and scope. The unified classification entry point rejects labels outside $\{0, 1\}$ on the entire input array. It also rejects negative or nonfinite weights and misaligned arrays; entirely zero weight yields unestimable metrics. Nonfinite scores are excluded from ROC, and missing energy is additionally excluded from energy-dependent statistics. An absent ROC class makes AUC unavailable; I can still be computed for estimable groups. If no category array is supplied, binary labels define the groups; missing entries in a supplied category array are rejected. The dependence routine filters nonfinite pairs; no correlation coefficient enters I . Constant scores produce identical histograms and $I = 1$. Constant conditioning energy likewise supplies no meaningful test of energy dependence.

The score in Equation 10 averages pairwise Jensen–Shannon divergences *before* taking a square root; it is not an average of Jensen–Shannon distances and not the multi-distribution Jensen–Shannon divergence. We use a descriptive benchmark name for this implemented transformation without claiming that it is a standard test or an original statistical construction.

B MJD Waveform Evidence for Energy-Aware Evaluation

Task, population, and selection. The local MJD release contains six test shards with 390,000 events, including 148,251 clean events. Clean means that all four reference PSD labels—low Aves, high

AvsE, DCR, and LQ—equal one; the formal task is clean versus non-clean, not low versus high energy. Calibrated energy is provided by `energy_label` in keV, and raw waveforms contain 3800 ADC samples (Arnquist et al., 2023b). We choose the detector/run pair with the most clean test events, breaking count ties by detector and run identifier. This gives detector 133, run 50928, with 410 events. Selection uses metadata and labels, before waveform inspection.

The lower and upper energy quartile boundaries are 226.871 and 471.248 keV. The corresponding cohorts contain 103 events each, covering 5.7–226.9 and 471.2–2616.4 keV; the middle group contains 204. For the event displays, we choose the event nearest each cohort’s median energy, with event identifier and row index breaking ties. The selected events have energies 157.975 and 1131.515 keV and identifiers 2407207 and 2585508. This rule does not inspect pulse shape or noise.

Actual waveform preprocessing. For raw waveform samples v_{it} , with $t = 0, \dots, 3799$, define

$$b_i = \frac{1}{200} \sum_{t=0}^{199} v_{it}, \quad \alpha_i = \max_t |v_{it} - b_i|, \quad u_{it} = \frac{v_{it} - b_i}{\alpha_i}. \quad (16)$$

The implementation reads float32 samples, computes the baseline mean in float64 and casts it to float32, then subtracts and normalizes in float32. Division is skipped when $\alpha_i = 0$; no selected event has zero amplitude. This is the common waveform preparation before the STFT or point representation, not an assertion that every classifier directly receives the plotted one-dimensional array. The time axis is sample index; no sampling interval is assumed.

We summarize residual baseline fluctuations by

$$r_i = \left(\frac{1}{200} \sum_{t=0}^{199} u_{it}^2 \right)^{1/2}. \quad (17)$$

Up to float32 rounding, r_i is baseline RMS in ADC counts divided by α_i . It is a descriptive input statistic, distinct from the benchmark’s energy independence score. The lowest/highest energy cohorts have median r_i values 0.006720 and 0.000889, a ratio of 7.56. Their absolute baseline RMS medians are much closer, 2.233 and 2.166 ADC counts. Across the 410 events, energy has Spearman correlation 0.993 with α_i , -0.012 with absolute baseline RMS, and -0.826 with r_i . These are single-cohort descriptive statistics, without uncertainty intervals or a claim that they hold for every detector/run. Neither correlation nor these event displays determines the strategy of a trained classifier.

Connection to the saved benchmark. Raw test labels and event IDs match the saved CNN evaluation population. Calibrated energies are recovered in float64 from the original test shards; the old adapter’s float32 conversion and clipping are not used. The 67 events above 3000 keV (66 non-clean and one clean) are excluded from energy evaluation before support estimation; they remain in the original coverage denominator. The common support is $[18.151, 2613.790]$ keV with 493 valid 5 keV bins, retaining 146,383 clean and 232,824 non-clean events. The CNN’s inclusive and matched AUCs are 0.971283 and 0.969851, respectively, while $I = 0.759637$. The small AUC difference does not establish energy-dominated classification. Matching controls the class energy mixture; the independence score asks a separate question about the predictions within each class.

C Energy-Only SuperNEMO Diagnostic

Population and operating point. This diagnostic uses all 196,608 simulated $0\nu\beta\beta$ and 262,144 ^{214}Bi events in the existing shared test partitions. It is separate from the trained $2\nu\beta\beta$ classification task. Energy is the reconstructed calorimeter sum $E_1 + E_2$ in keV, extracted once per event; the identical event-level fields repeated on tracker rows are not counted as separate observations. The released two-electron topology selection applies, with no additional energy cut or random sampling for the spectra. Class normalization removes differences in simulation sample counts; it does not estimate exposure-normalized physical rates.

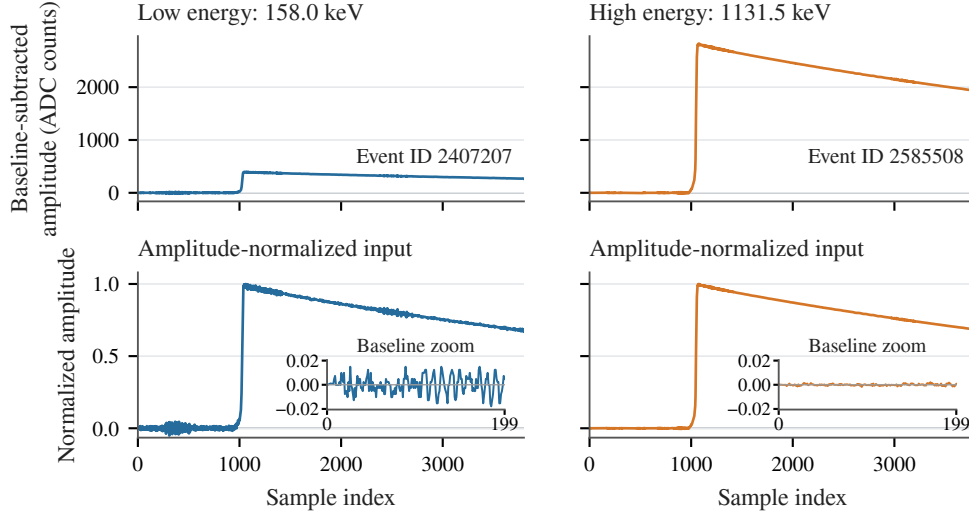
At the illustrative threshold $t = 2200$ keV, the rule $\hat{y}_i = 1\{E_i \geq t\}$ accepts 184,727 of 196,608 signal events and 26,793 of 262,144 background events. Hence

$$\text{TPR} = 0.939570, \quad \text{FPR} = 0.102207, \quad \text{BA} = \frac{1}{2}[\text{TPR} + (1 - \text{FPR})] = 0.918681, \quad (18)$$

MJD low- and high-energy waveforms

Clean test events; detector 133, run 50928; one event near each cohort median energy.

Low: 5.7–226.9 keV (n=103); high: 471.2–2616.4 keV (n=103)



Subtract the mean of samples 0–199; divide each event by its maximum absolute amplitude. Insets show samples 0–199. Row axes and insets share limits. Sampling interval is not assumed.

Figure 5: Detailed MJD event displays from the fixed detector/run cohort. Top: baseline-subtracted ADC amplitudes. Bottom: normalized waveforms before model-specific representation, with enlarged first-200-sample baselines. The two columns share vertical limits within each row, and both baseline insets use the same scale. Events were selected by energy alone using the stated median rule.

where BA is balanced accuracy, assigning equal importance to both classes. Figure 7 shows the operating-point trade-off. The threshold was specified for illustration rather than selected by maximizing test performance. Neither this cut nor the score $s_i = E_i$ requires training or event topology.

Matching and coverage. We apply the shared 600 regular 5 keV bins plus one overflow bin to this diagnostic pair, with class-wise 0.5% tail trimming, overlap matching, and at least 20 events of each class per bin. Common support remains $C = [1169.390, 2734.195]$ keV. The slice of the zero-anchored grid intersecting C has edges

$$(1165, 1170, 1175, \dots, 2730, 2735) \text{ keV}. \quad (19)$$

Events must also satisfy $E \in C$, so the boundary bins are partially occupied. Of 314 candidate bins, 283 pass the count requirement; sparse bins are excluded without merging. This retains 99,139 signal and 150,591 background events, or fractions 0.504247 and 0.574459, respectively. Both satisfy the minimum coverage of 0.5, with signal coverage close to the boundary. The matched result therefore describes the retained energy population, not the full signal peak. Reweighting gives effective sample sizes of approximately 34,127 and 64,317. The 50 keV display bins in Figures 3(a,b) and 6 do not set the matching resolution.

The inclusive ROC uses all original energies. The shared energy grid is assigned before support trimming, including 758 signal and 40 background events with $E > 3000$ keV in one overflow bin. All energies are finite and nonnegative. Here C lies below 3000 keV, so overflow events receive zero matched weight; they remain in the coverage denominators and in the separate independence calculation whenever the group/bin count rule passes. No energies are clipped. The trained $2\nu\beta\beta$ task uses the same evaluation rules on its own distinct test population.

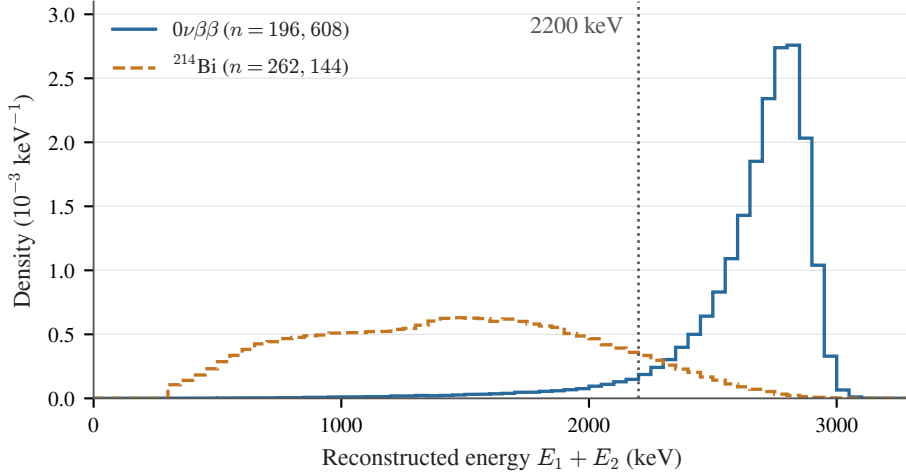


Figure 6: The motivating energy-spectrum difference: 196,608 simulated $0\nu\beta\beta$ and 262,144 ^{214}Bi events from shared SuperNEMO test partitions. The 50 keV display bins have unit area per class, so sample counts do not determine their relative heights. The released two-electron selection applies (Macko, 2026a); no additional energy cut is applied to the plotted spectra. The vertical line marks the illustrative 2200 keV threshold, not a test-optimized operating point.

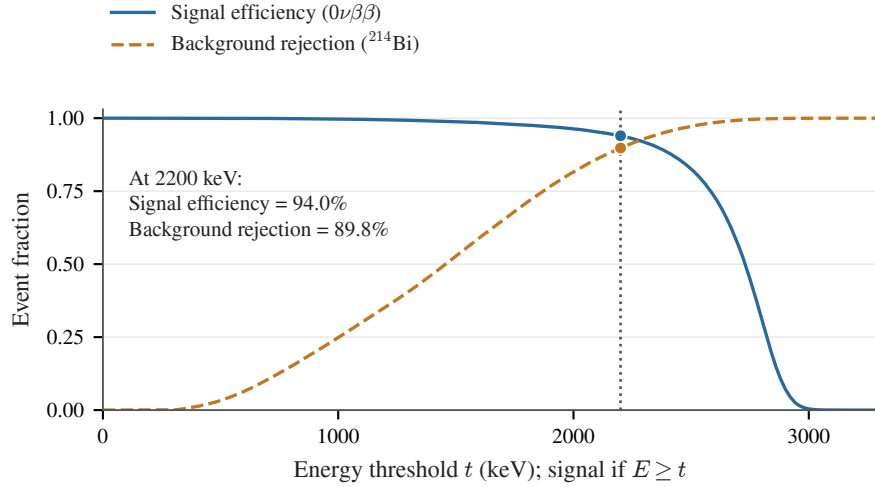


Figure 7: Signal efficiency and background rejection as an energy-only threshold varies in the same SuperNEMO $0\nu\beta\beta$ - ^{214}Bi test partitions. Curves use all events; the display samples thresholds every 10 keV. Markers identify the illustrative 2200 keV cut, with rates calculated directly from event energies.

Residual within-bin ordering. For the increasing score $s = E$, let q_k denote the shared matched mass of ordered energy bin k , and let A_k be the energy-only AUC within that bin, with half credit for ties. Cross-bin comparisons give

$$\text{AUC}_{\text{match}}(E) = \sum_{k>\ell} q_k q_\ell + \sum_k q_k^2 A_k = \frac{1}{2} + \sum_k q_k^2 \left(A_k - \frac{1}{2} \right). \quad (20)$$

Equal bin masses balance cross-bin pairs, while within-bin energy differences remain. This identity gives AUC 0.500024 for the fixed 5 keV matching bins; it does not redefine the metric as an average of per-bin AUCs. A score depending only on bin identity would instead have matched AUC exactly

Table 7: Energy-only AUC under successive evaluation conditions. Retained event counts are unweighted; only the last row uses matching weights. The populations and weights differ, so these rows are not competing models.

Evaluation condition	Signal events	Background events	AUC
Original test populations	196,608	262,144	0.970613
Common support, unit weights	99,595	172,757	0.934642
Retained 5 keV bins, unit weights	99,139	150,591	0.929715
Retained 5 keV bins, matched weights	99,139	150,591	0.500024

1/2. The independence calculation uses the same energy grid with original base weights and up to 20 score-quantile bins, yielding $I = 0.079210$ for $s = E$. Its class values are $I_{0\nu} = 0.082192$ and $I_{\text{Bi}} = 0.076972$. The 337 and 514 eligible group bins retain 195,169 signal and 261,873 background events, respectively, including their eligible overflow bins. The energy-only score still varies within each 5 keV bin; balanced bin masses therefore do not imply exact continuous-score AUC 1/2. These are deterministic diagnostic values, without bootstrap intervals or multiple-seed averages. They demonstrate the availability of an energy-only decision rule, not the internal strategy of any trained model or a measured failure under distribution shift.

D SuperNEMO Tracker Extent and Energy

For an event with h_i raw tracker-hit positions $\vec{r}_{ij}$ in millimeters, we measure spatial extent by

$$R_{g,i} = \left[\frac{1}{h_i} \sum_j \|\vec{r}_{ij} - \bar{\vec{r}}_i\|^2 \right]^{1/2}, \quad \bar{\vec{r}}_i = \frac{1}{h_i} \sum_j \vec{r}_{ij}. \quad (21)$$

This radius of gyration uses all stored tracker hit centers with equal weights; it is not a fitted track length. Reconstructed energy is the calorimeter sum $E_1 + E_2$, counted once per event. We retain all 262,144 ^{214}Bi and 196,608 $0\nu\beta\beta$ test events, with no missing energies or coordinate summaries.

Figure 8 compares complete populations. Each 100 keV energy column is normalized over all 20 mm extent bins. Columns with fewer than 20 events are masked, affecting one ^{214}Bi and 35 $0\nu\beta\beta$ events. Extents above 800 mm remain in the normalization but are outside the heatmap view; the cumulative plots include their full tails. These display bins are separate from EnergyBench matching bins.

To compare balanced cohorts, events are ranked by energy and then event ordinal within each physical class. We take the first and last quarter: 65,536 events per cohort for ^{214}Bi and 49,152 for $0\nu\beta\beta$. Exact energy ties are split by ordinal, not geometry. The ^{214}Bi quartile boundaries are 1003.0 and 1870.25 keV; the $0\nu\beta\beta$ boundaries are 2583.125 and 2821.5 keV. These relative energy cohorts are not common absolute windows across classes.

Median R_g decreases from 335.97 to 291.14 mm for the illustrative $0\nu\beta\beta$ sample and changes from 281.97 to 294.04 mm for ^{214}Bi . The distributions overlap substantially. Geometry can therefore contain energy-associated information, but these observations do not establish its use by a trained classifier or a universal monotonic size–energy relation. No specific physical or selection mechanism is inferred. The formal SuperNEMO benchmark remains $2\nu\beta\beta$ versus ^{214}Bi .

The current point adapters center events, quantize at 15 mm, and divide coordinates by a fixed 1000 mm scale; CNNs use a fixed 44 mm occupancy grid. They exclude calorimeter energies and do not normalize each event to unit spatial extent. Quantization and aggregation nevertheless change the input, so the raw extent statistic is not a claim about the learned representation.

E Non-Transformer Architectures and Run Protocols

E.1 Evidence Supporting the Four-Model Comparison

Unified reevaluation of earlier NEXT checkpoints provides retrospective context for the representative architectures. In the alternative-model batch, the multi-view CNN has matched AUC 0.955804,

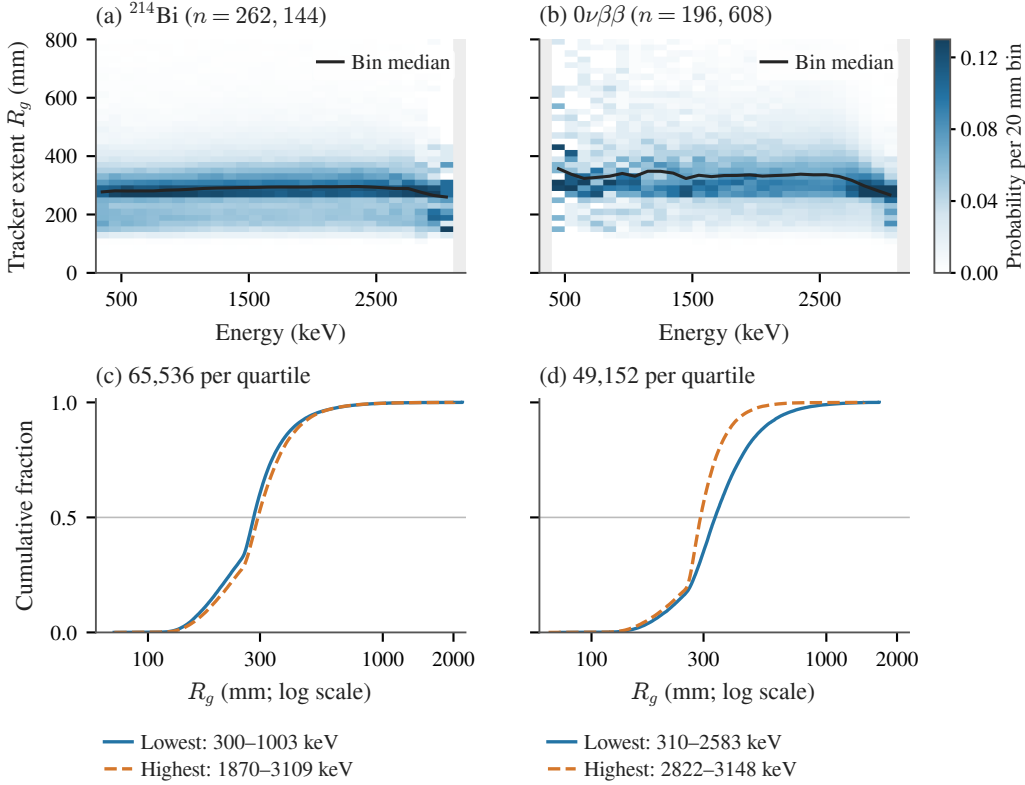


Figure 8: SuperNEMO tracker extent versus reconstructed energy. Top: conditional distributions using 100 keV by 20 mm bins; each energy column is normalized over all extents. Black curves show medians; gray columns contain fewer than 20 events. Extents above 800 mm (0.99% of ^{214}Bi and 1.10% of $0\nu\beta\beta$) are excluded from view only. Bottom: complete cumulative distributions for the lowest and highest energy-ranked quartiles within each class, on a logarithmic extent axis. Cohort sizes and energy ranges are given in the panels.

compared with 0.947782 for the multiscale CNN and 0.928273 for the dense 3D CNN; Static GINE leads the graph candidates in that batch at 0.981008. In the subsequent candidate batch, BiGRU exceeds the tested dilated TCN (0.933864 versus 0.926085). PointMamba-lite, the only recorded state-space candidate, has matched AUC 0.925386 and $I = 0.971387$; its inclusion broadens the architectural comparison rather than following an overall accuracy ranking.

These are held-out test comparisons on 115,499 events from earlier exploratory batches. They are distinct from the 116,549-event reevaluation used for the available NEXT classic results in Table 1. PointMamba-lite is included in Table 10. The original exploratory batches used different evaluator-code fingerprints; the numbers above now use the same regular-plus-overflow protocol as the current classic results while retaining the earlier checkpoints and test set. Their distinct test population is not pooled with the retained campaign. The four search groups are recorded in the model-family presentation and are reproduced in Table 10. They organize the candidate inventory; the surviving records do not document a prospective validation-based rule selecting one winner from each group. The retained architectures span three of the four groups, including two distinct mechanisms from the sequence/state-space group. The structural rationale and early test comparisons therefore support the comparison set without establishing a recovered formal selection rule.

Parameter-count provenance. Parameter counts in Table 10 are the saved counts of trainable neural parameters, checked against the corresponding run summaries and reevaluated metrics. Neural parameter count is inapplicable to XGBoost. The earlier NEXT PointMamba-lite run summary records 316,993 trainable parameters, consistent with Benchmarking table.xlsx; this is displayed as

Table 8: Input adaptations for the four non-Transformer architectures. GINE, BiGRU, and PointMamba-lite share each dataset’s point representation. Model dimensions and representation details are given in Appendix E.

Dataset	MV-CNN input	Graph and sequence input
NEXT	Three normalized spatial energy projections	Centered 15 mm voxels; energy fraction and log hit count
SuperNEMO	Three binary tracker projections	Centered 15 mm tracker cells; occupancy fraction and log hit count
MJD	Three-resolution waveform STFT	512 pooled waveform segments with mean and RMS features
EXO-200	STFT of serialized 226×300 waveform	Same waveform construction after channel-major serialization

0.3170 M in Table 10 and 317 K in Table 1. Its retained-campaign predictions remain unavailable. Figure 4 includes only the retained-campaign configurations with available predictions.

Spatial representations. NEXT MV-CNN uses three 128×128 projections with 30 mm bins. For point-based models, hits are centered by the complete-event energy centroid and merged into 15 mm cells. Voxel centers are recentered and divided by 1,000 mm; the two features are the voxel’s fraction of complete-event energy and $\log(1+n)$, where n is its hit count. Events exceeding 512 occupied cells retain the highest-energy cells with deterministic tie breaking, without renormalizing the retained energy fractions. SuperNEMO instead uses tracker coordinates: its CNN projects binary occupancy on 128×128 grids with 44 mm bins. Its point adapter centers the tracker hits geometrically and merges 15 mm cells, replacing the unavailable hit-energy fractions with occupancy fractions. It raises an error above 512 cells instead of truncating. The non-Transformer SuperNEMO inputs exclude E_1 , E_2 , and tracker drift radii. These input choices restrict direct energy access but do not establish independence from energy.

Waveform representations. MJD subtracts the mean of the first 200 waveform samples and divides by the maximum absolute event amplitude when nonzero. EXO-200 subtracts a baseline separately for each channel using its first 200 time samples, divides by the event-wide maximum absolute amplitude, and serializes the 226×300 array in channel-major order. For these datasets, MV-CNN uses Hann-window STFTs with FFT sizes (64, 128, 256) and hops (16, 32, 64), followed by a $\log(1 + |\text{STFT}|)$ transform and resizing to 128×128 . The point adapter uses adaptive pooling to form 512 segment means and root-mean-square amplitudes. Its three coordinates are normalized time, amplitude-normalized segment mean, and the first difference of that normalized mean; its two features are the segment mean and RMS. These are pulse-shape coordinates, not calibrated detector positions.

Backbone details. MV-CNN has four residual stages with two blocks each, base width 16, and a 256-dimensional fusion layer. GINE uses hidden width 128, five message-passing layers, $k = 12$ neighbors, and a 160-dimensional classifier. BiGRU uses a 96-dimensional input embedding, two bidirectional layers with hidden size 128, and a 256-dimensional classifier. The two Hilbert orders quantize each event’s point bounding box to 10 bits per axis; the second swaps the first two axes before applying the same ordering. PointMamba-lite uses three blocks with model width 128, inner width 192, state dimension 16, and 32-position scan chunks. Its guarded PyTorch scan is not the official fused Mamba implementation. The backbones use dropout 0.1. MJD GINE has four output logits z_{ij} , one per reference label, and uses

$$q_i = \prod_{j=1}^4 \sigma(z_{ij}), \quad s_i = \log q_i - \log(1 - q_i), \quad (22)$$

where σ is the logistic sigmoid and s_i is its evaluated clean-event score. This product is the implemented score construction; it is not a demonstrated factorization of the labels’ joint conditional distribution. The computation uses summed log-sigmoids and clamps the log probability below zero for numerical stability. EXO-200 uses the same GINE backbone with one output.

Table 9: Specified common protocol versus recorded training settings for the audited non-Transformer models. Every listed configuration has an upper limit of 50 epochs, weight decay 10^{-4} , gradient-norm limit 1, and seed 42. NEXT settings refer to the three available retained-campaign runs. “Other datasets” means EXO-200, MJD, and SuperNEMO.

Scope	Architecture	Batch	Initial learning rate	Patience
Specified protocol	Neural models	64	5×10^{-4}	5
NEXT recorded	MV-CNN, GINE, BiGRU	64	5×10^{-4}	5
Other datasets	MV-CNN	16	10^{-3}	12
Other datasets	GINE, BiGRU	16	5×10^{-4}	12
Other datasets	PointMamba-lite	4	3×10^{-4}	12

Run completion and randomness. The audited non-Transformer classification loaders retain the sampled class proportions; their binary losses do not use class weights, and these paths contain no synthetic data augmentation. NEXT shuffles training slices and buffered events, waveform loaders shuffle training events, and SuperNEMO records proportional sampling without replacement. Input normalization and ordering are dataset-specific preprocessing rather than energy matching of training events.

In the NEXT campaign reported here, MV-CNN, GINE, and BiGRU finish after 19, 36, and 26 epochs, selecting epochs 14, 31, and 21, respectively, by validation inclusive AUC. The loader batch size is 64; the wrapper evaluates these backbones in chunks of 16, concatenates their logits, and applies one batch loss and optimizer step. The corresponding run summaries record Python 3.11.15, PyTorch 2.11.0+cu128, an NVIDIA GeForce RTX 5090, and bfloat16 mixed precision. This environment evidence applies to those runs rather than automatically to every dataset. The audited non-Transformer training paths add neither class-weighted losses nor data augmentation; the current SuperNEMO stream retains the source class proportions without replacement.

For SuperNEMO, the workbook records 21, 50, 22, and 9 training epochs for MV-CNN, GINE, BiGRU, and PointMamba-lite, with selected epochs 16, 43, 17, and 4, respectively. It records 47 epochs and selected epoch 43 for MJD PointMamba-lite. Earlier logs were incomplete; the current results and epoch counts follow the workbook, whose evaluation status does not establish training completion. The audited non-Transformer configurations set seed 42 and disable enforced deterministic algorithms. Mixed precision is enabled in these configurations except for MJD BiGRU. Seeds therefore do not imply bitwise reproducibility.

Screening scope. The current NEXT registry contains three legacy CNNs and twenty alternative classifiers, including point, graph, sequence, state-space, sparse-convolution, hybrid, mixer, persistence, and boosted-tree approaches. A classification entry need not correspond to a completed retained-campaign run: PointMamba-lite has an earlier exploratory result but remains unavailable in that campaign. The search taxonomy groups the three legacy projection CNNs together with projection, patch-mixer, dense-voxel, and sparse-voxel alternatives. Point-set and graph methods form separate groups; ordered sequence methods, the state-space model, topological classifiers, and the CNN-GNN hybrid form the fourth. This representation-based organization differs from the finer model-family labels in the run registry. Checkpoint selection is documented within completed runs, while a prospective rule for choosing the cross-dataset architectures is not. The boosted-tree classifier in the extended NEXT inventory uses up to 500 estimators, learning rate 0.04, maximum depth 4, subsampling 0.85, column subsampling 0.9, early-stopping rounds 12, and random seed 42. Its estimator settings, rather than the neural epoch, optimizer, and clipping fields in the generic run configuration, determine its training procedure.

F Extended NEXT Classification Results

Table 10: NEXT classification candidates reevaluated with 600 regular 5 keV bins on 0–3000 keV plus one overflow bin above 3000 keV. Params (M) counts trainable neural parameters in millions. A dash denotes an inapplicable parameter count. [†]Architecture instantiated across datasets; no new model is fitted.

Model	Params (M)	Inclusive AUC $\uparrow$	Matched AUC $\uparrow$	I $\uparrow$
<i>Projection and voxel models</i>				
Two-conv CNN	0.0014	0.953280	0.952992	0.966472
Energy-skip CNN	0.0014	0.923394	0.922735	0.967952
Residual spatial CNN	1.2288	0.979030	0.978966	0.969531
Multi-view CNN [†]	0.8077	0.976636	0.976587	0.970993
Multi-scale CNN	0.8659	0.979270	0.979046	0.973145
Dense 3D ResNet	0.6884	0.984115	0.983956	0.971101
Projection MLP mixer	0.5391	0.950352	0.950086	0.972549
Submanifold ResNet	0.2982	0.991574	0.991414	0.967824
<i>Point-set models</i>				
DeepSets	0.0756	0.969660	0.969459	0.972602
PointNet++	0.1645	0.985459	0.985354	0.972546
PointMLP	0.4435	0.994033	0.993814	0.967686
Rigid KPConv	0.3910	0.986654	0.986341	0.969283
<i>Graph models</i>				
Static GINE [†]	0.4799	0.994801	0.994625	0.970265
ParticleNet-lite	0.1505	0.992161	0.991971	0.968044
EGNN	0.6405	0.890742	0.889878	0.968064
GravNet	0.2936	0.991316	0.991070	0.969637
DimeNet-lite	0.1696	0.992939	0.992716	0.968001
<i>Sequence, state-space, topology and hybrid models</i>				
BiGRU [†]	0.7340	0.979563	0.979377	0.971051
Dilated TCN	0.6945	0.973450	0.973183	0.974533
PointMamba-lite [†]	0.3170	0.925742	0.925386	0.971387
Persistence/PersLay	0.0633	0.918258	0.917657	0.968522
Topology XGBoost	—	0.937279	0.936938	0.970396
CNN-GNN	0.3420	0.991417	0.991260	0.970276

G Transformer Representation Details

Table 11 records the concrete dataset-specific implementations of the three tokenization families used in Section 3.2. Here, d_f is the token content width and d_p is the coordinate width. A token count preceded by $\leq$ denotes a maximum sequence length; shorter events are masked and padded only for batching.

NEXT and SuperNEMO use three-dimensional detector position coordinates. MJD uses normalized time together with normalized waveform level and local temporal change. EXO-200 uses normalized time, within-block channel position, detector side, and indicators for the U-wire, V-wire, and APD sensor families.

Table 11: Dataset-specific Transformer tokenization configurations. Internal feature padding, such as the channel padding used by EXO-200 raw patches, does not introduce additional Transformer tokens.

Dataset	Family	Tokens	d_f/d_p	Construction
NEXT	Entity	≤ 512	2/3	Individual detector hits selected under a fixed token cap. Token content describes the local hit measurement, while centered three-dimensional coordinates provide position.
NEXT	Region	≤ 512	2/3	Hits are aggregated within occupied three-dimensional voxels of width 15 detector-coordinate units. If necessary, the most occupied voxels are retained under the token cap.
NEXT	Summary	≤ 512	4/3	Hits are ordered using a three-dimensional Morton ordering and partitioned into balanced spatial groups. Each token summarizes the measurements within one group.
MJD	Entity	500	2/3	The length-3800 waveform is represented by a deterministic mixture of uniformly spaced and high-importance samples. Local context spans nine waveform samples.
MJD	Region	500	20/3	The waveform is divided into 512 non-overlapping patches containing 20 raw amplitudes each.
MJD	Summary	500	2/3	The same 512 temporal regions are represented by their segment mean and root-mean-square amplitude.
EXO-200	Entity	504	2/6	Entities are selected independently within each of six sensor blocks using equal proportions of uniform and waveform-salient samples. Each token stores amplitude and nine-sample local RMS.
EXO-200	Region	504	150/6	Six sensor blocks are divided into seven channel regions and twelve time regions. Each token retains up to 6×25 raw waveform values; smaller channel regions are padded only within the token feature.
EXO-200	Summary	504	4/6	The same sensor-temporal regions are described by mean, RMS, maximum absolute amplitude, and the difference between the second- and first-half means.
SuperNEMO	Entity	≤ 512	4/3	Individual tracker hits form tokens under a deterministic event-level cap, with centered three-dimensional detector coordinates.
SuperNEMO	Region	≤ 512	8/3	Hits are aggregated into occupied cubic spatial regions with width 88 mm. Occupancy-based selection is used if an event exceeds the token cap.
SuperNEMO	Summary	≤ 16	8/3	Hits are Morton ordered and divided into at most sixteen balanced spatial groups, each represented by aggregated event statistics.